

Lesson 15: Movies Analyses Dashboard with DAX

Topic: Movies Analyses Dashboard with DAX

Prerequisites: All data is provided in *Movies.xlsx*

Step 1: Load Data

- Import Movies.xlsx into **Power BI Desktop**.
 - Inspect all columns for missing or incorrect data.
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Step 2: Data Cleaning

Handle Missing Values

- Remove or **impute missing values** in Budget and Box Office columns.
 - Option 1: Replace blanks with **0**.
 - Budget_Clean = IF(ISBLANK(Movies[Budget]), 0, Movies[Budget])
 - BoxOffice_Clean = IF(ISBLANK(Movies[Box Office]), 0, Movies[Box Office])
 - Option 2: Replace with **median value** (optional).

Ensure Correct Data Types

- Check all columns:
 - **Release Date** → *Date*
 - **Budget** and **Box Office** → *Currency or Decimal Number*
 - **Run Time** → *Whole Number*
 - **Genre, Director, Country** → *Text*
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Step 3: Create a Date Table

Use Release Date to create a **Date Table** with DAX:

DateTable =

ADDCOLUMNS(

 CALENDARAUTO(),

 "Year", YEAR([Date]),

 "Month", FORMAT([Date], "MMMM"),

```
"MonthNumber", MONTH([Date]),  
"Quarter", "Q" & FORMAT([Date], "Q")  
)
```

- In **Model View**, create a relationship:
 - Movies[Release Date] → DataTable[Date]
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Step 4: Create Calculated Columns

1. Profit

Profit = Movies[BoxOffice_Clean] - Movies[Budget_Clean]

2. Run Time Category

Run Time Category =

```
SWITCH(  
    TRUE(),  
    Movies[Run Time] < 90, "Short",  
    Movies[Run Time] >= 90 && Movies[Run Time] < 120, "Medium",  
    Movies[Run Time] >= 120, "Long",  
    "Unknown"  
)
```

Step 5: Create DAX Measures

1. Total Box Office

Total Box Office = SUM(Movies[BoxOffice_Clean])

2. Average Budget

Average Budget = AVERAGE(Movies[Budget_Clean])

3. Average Margin

Average Margin = AVERAGE(Movies[Profit])

4. Total Movies with Oscars

Total Movies with Oscars = COUNTROWS(FILTER(Movies, Movies[Oscar Wins] > 0))

5. Top Genre by Box Office

Top Genre by Box Office =

VAR GenreSales =

```
SUMMARIZE(  
    Movies,  
    Movies[Genre],  
    "TotalBoxOffice", SUM(Movies[BoxOffice_Clean])  
)
```

RETURN

TOPN(1, GenreSales, [TotalBoxOffice], DESC)

6. Year-over-Year (YoY) Box Office Growth

YoY Box Office Growth =

VAR PrevYear =

```
CALCULATE([Total Box Office], DATEADD(DateTable[Date], -1, YEAR))
```

RETURN

```
DIVIDE([Total Box Office] - PrevYear, PrevYear)
```

7. Average Nominations per Director

Average Nominations per Director =

```
AVERAGEX(  
    VALUES(Movies[Director]),  
    CALCULATE(SUM(Movies[Nominations]))  
)
```

Step 6: Visualization Requirements

Page 1: Overview Dashboard

Recommended Visuals:

- **Cards:**
 - Total Box Office
 - Profit Margin
 - Movies with Oscars

- **Bar Chart:** Total Box Office by Genre (stacked by Certificate)
 - **Line Chart:** Box Office trend by Release Year
 - **Slicer:** Country and Release Date (range)
 - **KPI:** YoY Box Office Growth (target > 0%)
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Page 2: Director Analysis

Recommended Visuals:

- **Treemap:** Budget by Director (size by Budget, color by Oscar Wins)
 - **Table:** Director, Total Nominations, Total Oscars, Avg Nominations per Director
 - **Slicer:** Filter by Genre
 - **Donut Chart:** Run Time Category distribution per selected Director
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Page 3: Genre and Country Insights

Recommended Visuals:

- **Matrix:** Genre vs. Country (Total Box Office as values)
 - **Pie Chart:** Share of Box Office by Certificate
 - **Word Cloud:** Genre (requires custom visual from Marketplace)
 - **Slicer:** Run Time Category
 - Apply **Conditional Formatting** in Matrix for color scale by Box Office.
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Final Notes

- ✓ Ensure all relationships are active and correctly set in **Model View**.
- ✓ Format visuals using consistent colors and layout.
- ✓ Use **Bookmarks** or **Navigation Buttons** to switch between dashboard pages.